

Model	LR	Epoch	Batch Size	Cutoff Length	Template	Scheduler	Warmup Ratio	Weight
LLaMA1-7B _{9k}	2e-5	3	128	2048	vicuna	cosine	0.1	FP32
LLaMA1-7B _{39k}	2e-5	3	128	2048	vicuna	cosine	0.1	FP32
LLaMA1-7B _{52k}	2e-5	3	128	2048	vicuna	cosine	0.1	FP32

Table 7: Configuration of training hyperparameters used in Figure 1 (following (Zhou et al. 2024), LLaMA1-7B_{9k} denotes LLaMA1-7B language model finetuned on 9K subset of Alpaca 52K with threshold $\tau = 4.5$).

Model	LR	Epoch	Batch Size	Cutoff Length	Template	Scheduler	Warmup Ratio	Weight
$\phi_{quality}$	2e-5	3	5	2048	-	linear	0	FP32
$\phi_{complexity}$	2e-5	3	5	1024	-	linear	0	FP32
Mistral-7B-v0.1	2e-5	6	512	2048	vicuna	cosine	0.1	FP32
LLaMA1-13B	2e-5	6	512	2048	vicuna	cosine	0.1	FP16
LLaMA2-13B	2e-5	6	512	2048	vicuna	cosine	0.1	FP16

Table 8: Configuration of training hyperparameters (following (Lu et al. 2023; Liu et al. 2023)).

A Appendix

A.1 Instructions

The part enclosed by curly braces is the field that needs to be filled in.

A.2 Training Details

We show the training hyperparameters in Table 7 and Table 8. We utilize four/eight NVIDIA Tesla V100 GPUs to train 7B/13B models. In order to reduce the impact of different training parameters, we mainly set the training parameters based on the baselines paper (Tunstall et al. 2023; Lu et al. 2023; Liu et al. 2023; Zhou et al. 2024). Due to resource limitation, we maintain total batch size to 512, which is the result of multiplying number of GPUs, batch size on per device, and gradient accumulation steps. Specifically, we set batch size on per device to 4, and gradient accumulation steps to 2. We release our code anonymously: <https://anonymous.4open.science/r/DataAgent-1F2E>.

A.3 Data Pool

A.4 Refine Times Effect

Table 11 summarizes the evaluation results on traditional LLM benchmarks. LLaMA1-13B and LLaMA2-13B both achieve the best average results when refine 1 time, and achieve the second best results when refine 3 times. When applying refine distillation, the average score increased by 1.20 and 1.42, respectively, which also demonstrates the effectiveness of refine distillation.

Dataset	Data Size
ShareGPT (Chiang et al. 2023)	58K
UltraChat (Ding et al. 2023)	105K
WizardLM (Xu et al. 2023)	143K

Table 9: Statistic information of data pool.

Test Dataset	Data Size
Koala (Geng et al. 2023)	180
Vicuna (Chiang et al. 2023)	80
WizardLM (Xu et al. 2023)	218
Self-Instruct (Wang et al. 2022b)	252

Table 10: Statistic information of four test datasets used in the win rate experiments (Figure 1).

#Refine	ARC	HellaSwag	MMLU	TruthfulQA	Average
Open-sourced Models based on LLaMA1-13B					
0	58.36	81.80	47.55	52.75	60.11
1	58.62	81.84	48.03	56.73	61.31
2	58.79	82.09	47.49	55.95	61.08
3	59.22	82.13	47.81	55.83	<u>61.25</u>
4	59.13	82.18	48.44	55.05	61.20
5	59.22	82.14	48.16	53.60	60.78
Open-sourced Models based on LLaMA2-13B					
0	59.22	80.73	53.29	50.29	60.88
1	59.90	81.83	54.05	53.44	62.30
2	60.32	81.80	53.08	50.53	61.43
3	62.12	81.31	53.36	51.36	<u>62.04</u>
4	62.37	81.24	53.51	50.55	61.92
5	60.92	81.54	54.09	50.97	61.88

Table 11: Results on the traditional LLM benchmarks with different refine times.

A.5 Auto-grader Instruction

System Prompt:

We would like to request your feedback on the performance of AI assistant in response to the instruction and the given input displayed following.

Instruction: {instruction}

Input: {input}

Response: {response}

User Prompt:

Please rate according to the accuracy of the response to the instruction and the input. Each assistant receives a score on a scale of 0 to 5, where a higher score indicates higher level of the [accuracy]. Please first output a single line containing the value indicating the scores. In the subsequent line, please provide a comprehensive explanation of your evaluation, avoiding any potential bias.

Following (Chen et al. 2023b), we use the instruction A.5 to prompt gpt-3.5-turbo-0301 to rate the Alpaca-52K dataset.

A.6 Adding Constraints Instruction

I want you act as a Prompt Rewriter.
Your objective is to rewrite a given prompt into a more complex version to make those famous AI systems (e.g., chatgpt and GPT4) a bit harder to handle.
But the rewritten prompt must be reasonable and must be understood and responded by humans.
Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt#. Also, please do not omit the input in #Given Prompt#.
You SHOULD complicate the given prompt using the following method:
Please add one more constraints/requirements into #Given Prompt#
You should try your best not to make the #Rewritten Prompt# become verbose, #Rewritten Prompt# can only add 10 to 20 words into #Given Prompt#.
'#Given Prompt#', '#Rewritten Prompt#', 'given prompt' and 'rewritten prompt' are not allowed to appear in #Rewritten Prompt#

#Given Prompt#:
{given_prompt}
#Created Prompt#:

A.7 Creating Deeper Instruction

I want you act as a Prompt Rewriter.
Your objective is to rewrite a given prompt into a more complex version to make those famous AI systems (e.g., chatgpt and GPT4) a bit harder to handle.
But the rewritten prompt must be reasonable and must be understood and responded by humans.
Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt#. Also, please do not omit the input in #Given Prompt#.
You SHOULD complicate the given prompt using the following method:
If #Given Prompt# contains inquiries about certain issues, the depth and breadth of the inquiry can be increased.
You should try your best not to make the #Rewritten Prompt# become verbose, #Rewritten Prompt# can only add 10 to 20 words into #Given Prompt#.
'#Given Prompt#', '#Rewritten Prompt#', 'given prompt' and 'rewritten prompt' are not allowed to appear in #Rewritten Prompt#

#Given Prompt#:
{given_prompt}
#Created Prompt#:

A.8 Creating Concretizing Instruction

I want you act as a Prompt Rewriter.
Your objective is to rewrite a given prompt into a more complex version to make those famous AI systems (e.g., chatgpt and GPT4) a bit harder to handle.
But the rewritten prompt must be reasonable and must be understood and responded by humans.
Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt#. Also, please do not omit the input in #Given Prompt#.
You SHOULD complicate the given prompt using the following method:
Please replace general concepts with more specific concepts.
You should try your best not to make the #Rewritten Prompt# become verbose, #Rewritten Prompt# can only add 10 to 20 words into #Given Prompt#.
'#Given Prompt#', '#Rewritten Prompt#', 'given prompt' and 'rewritten prompt' are not allowed to appear in #Rewritten Prompt#

#Given Prompt#:
{given_prompt}
#Created Prompt#:

A.9 Increase Reasoning Instruction

I want you act as a Prompt Rewriter.

Your objective is to rewrite a given prompt into a more complex version to make those famous AI systems (e.g., chatgpt and GPT4) a bit harder to handle.

But the rewritten prompt must be reasonable and must be understood and responded by humans.

Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt#. Also, please do not omit the input in #Given Prompt#.

You SHOULD complicate the given prompt using the following method:

If #Given Prompt# can be solved with just a few simple thinking processes, you can rewrite it to explicitly request multiple-step reasoning.

You should try your best not to make the #Rewritten Prompt# become verbose, #Rewritten Prompt# can only add 10 to 20 words into #Given Prompt#.

'#Given Prompt#', '#Rewritten Prompt#', 'given prompt' and 'rewritten prompt' are not allowed to appear in #Rewritten Prompt#

#Given Prompt#:

{given_prompt}

#Created Prompt#:

A.10 Increase the Brand Instruction

I want you act as a Prompt Creator.

Your goal is to draw inspiration from the #Given Prompt# to create a brand new prompt.

This new prompt should belong to the same domain as the #Given Prompt# but be even more rare.

The LENGTH and complexity of the #Created Prompt# should be similar to that of the #Given Prompt#.

The #Created Prompt# must be reasonable and must be understood and responded by humans.

'#Given Prompt#', '#Created Prompt#', 'given prompt' and 'created prompt' are not allowed to appear in #Created Prompt#.

#Given Prompt#:

{given_prompt}

#Created Prompt#:

A.11 Augmenting Response with Relevance Instruction

I want you to act as a Response Rewriter.

Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.

But the rewritten response must be reasonable and must be understood by humans.

Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#. Also, please do not omit the input in #Given Prompt#.

You Should enhance the quality of the response using the following method:

Please make the Response more **relevant** to the user.

You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#.

'#Given Response#', '#Rewritten Response#', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:

{given_prompt}

#Given Response#

{given_response}

#Rewritten Response#:

A.12 Enhancing Response with Helpfulness Instruction

I want you to act as a Response Rewriter.

Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.

But the rewritten response must be reasonable and must be understood by humans.

Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#.

Also, please do not omit the input in #Given Prompt#.

You Should enhance the quality of the response using the following method:

Please make the Response more **helpful** to the user.

You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#.

'#Given Response#', '#Rewritten Response#', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:

{given_prompt}

#Given Response#

{given_response}

#Rewritten Response#:

A.13 Enhancing Response with Enriching Depth Instruction

I want you to act as a Response Rewriter.

Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.

But the rewritten response must be reasonable and must be understood by humans.

Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#.

Also, please do not omit the input in #Given Prompt#.

You Should enhance the quality of the response using the following method:

Please make the Response more **in-depth**.

You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#.

'#Given Response#', '#Rewritten Response#', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:

{given_prompt}

#Given Response#

{given_response}

#Rewritten Response#:

A.14 Enhancing Response with Fostering Creativity Instruction

I want you to act as a Response Rewriter.
Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.
But the rewritten response must be reasonable and must be understood by humans.
Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#.
Also, please do not omit the input in #Given Prompt#.
You Should enhance the quality of the response using the following method:
Please **increase the creativity** of the response.
You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#.
'#Given Response#', '#Rewritten Response#', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:
{given_prompt}
#Given Response#
{given_response}
#Rewritten Response#:

A.15 Enhancing Response with Supplying Additional Details Instruction

I want you to act as a Response Rewriter.
Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.
But the rewritten response must be reasonable and must be understood by humans.
Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#.
Also, please do not omit the input in #Given Prompt#.
You Should enhance the quality of the response using the following method:
Please **increase the detail level** of Response.
You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#.
'#Given Response#', '#Rewritten Response#', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:
{given_prompt}
#Given Response#
{given_response}
#Rewritten Response#:

A.16 Instruction Ranking Template

You are RankGPT, an intelligent assistant that can rank instructions based on their instruction-following difficulty and complexity.
Ranking the following instructions provided by different users according to the instruction-following difficulty and complexity.
The following are {num} instructions, each indicated by number identifier [].
The user's instructions are:
{instructions_at_all_evolutionary_stage}
You will rank the {num} instructions above based on the instruction-following difficulty and complexity.
The instructions will be listed in descending order using identifiers, and the most relevant response should be listed first, and the output format should be [] > [] > etc, e.g., [1] > [2] > etc.
The ranking results of the {num} responses (only identifiers) is:

Algorithm 1: Data Selection

```
1: Input: The data pool  $X$ , select data size  $m$ , LLM  $\theta$ , threshold  $\tau$ , decay rate  $\gamma$ 
2: Output: The selected subset  $S^m$ 
3: Sorting  $X$  with the total score  $s_{total} = \sum_{t=0}^T (s_{comp} * s_{qual} + s_{qual} * s_{know})e^{-\gamma t}$ ;
4: Obtaining the sorted pool  $X^*$ ;
5: Initialize selected dataset  $S^m = \{x_0^*\}$ 
6: for Each sample  $x \in X^*$  do
7:   //  $d(x, S^m)$  denotes the distance between  $x$  and its nearest neighbor in  $S^m$ 
8:   if  $d(x, S^m) < \tau$  then
9:      $S^m \leftarrow S^m \cup \{x\}$ 
10:  else
11:    Continue
12:  end if
13:   $X \leftarrow X \setminus \{x\}$ 
14:  if  $|S^m|$  equals to  $m$  then
15:    Break
16:  end if
17: end for
```

A.17 Response Ranking Template

You are RankGPT, an intelligent assistant that can rank responses based on their questions. Your evaluation should consider factors such as helpfulness, relevance, accuracy, depth, creativity, and level of detail of the response.

The following are num responses, each indicated by number identifier [].

You can rank them based on their relevance to the question: {question}

{responses_at_all_evolutionary_stage}

The user's question is:

{question}

You will rank the num responses above based on their relevance to the question.

The responses will be listed in descending order using identifiers, and the most relevant response should be listed first, and the output format should be [] > [] > etc, e.g., [1] > [2] > etc.

The ranking results of the num responses (only identifiers) is:

A.18 Data Selection Algorithm

A.19 Score Distribution

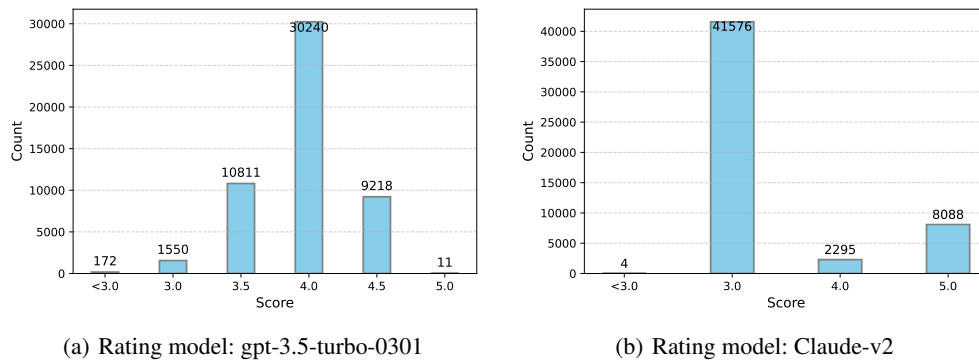
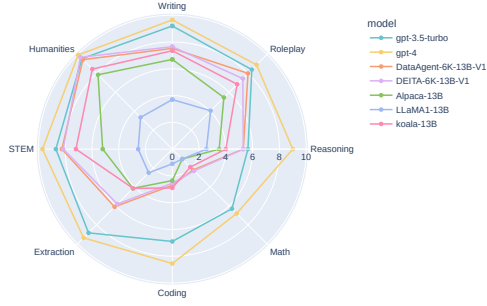


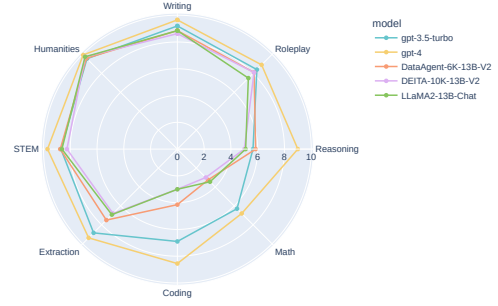
Figure 6: AlpaGasus score distribution on Alpaca dataset with prompt A.5.

Following AlpaGasus (Chen et al. 2023b), we prompt gpt-3.5-turbo-0301 (Ouyang et al. 2022) and Claude-v2 (Bai et al. 2022) to rate each data point in Alpaca dataset. We report the score distribution in Figure 6. We can find that gpt-3.5-turbo-0301 and Claude-v2 tend to give similar scores. Specifically, gpt-3.5-turbo-0301 gives a 4.0 score to 58.15% of the samples, and Claude-v2 gives a score of 3.0 to 80.01% of the samples, which is difficult for subsequent data sorting.

A.20 Detailed Scores in MT-Bench



(a) Baselines based on LLaMA1-13B



(b) Baselines based on LLaMA2-13B

Figure 7: Detailed scores across 8 subtasks in MT-Bench (DataAgent-6K-13B-V1 denotes LLaMA1-13B fine-tuned on 6K dataset filtered by DataAgent, DataAgent-6K-13B-V2 denotes LLaMA2-13B fine-tuned on 6K dataset filtered by DataAgent, DEITA-6K-13B-V1 denotes LLaMA1-13B fine-tuned on 6K dataset filtered by DEITA, and DEITA-6K-13B-V2 denotes LLaMA2-13B fine-tuned on 6K dataset filtered by DEITA).

As shown in Figure 7, we plot the standard radar plot of the 8 subtasks on MT-Bench. It clearly demonstrates that DataAgent achieves the best average performance, except for gpt-3.5-turbo and gpt-4. Specifically, DataAgent-6K-13B-V2 presents similar performance on STEM and Humanities, slightly weaker abilities on Writing and Roleplay, slightly better performance than gpt-3.5-turbo on Reasoning. However, in terms of performance on Coding and Math categories, there is still a significant gap compared to cutting-edge models.